Nationwide Survey of the Undergraduate Physical Chemistry Course

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ABSTRACT:

A nationwide survey of the undergraduate physical chemistry course was conducted to investigate the depth and breadth of content that is covered, how content is delivered, how student understanding is assessed, and the experiences and beliefs of instructors. The survey was administered to instructors of physical chemistry (*N* = 331) at American Chemical Society certified colleges and universities in the United States. The goal of this preliminary study is to provide an awareness of the current state of physical chemistry courses across the United States. Information gained from this study may inform the development of new teaching practices, curriculum, assessments, and resources for physical chemistry education, as well as encourage instructors of physical chemistry and researches of chemical education to work together to improve student understanding of physical chemistry.

RESEARCH QUESTIONS:

- What is the current state of physical chemistry courses across the country?
- What is the relationship between the experiences and beliefs of instructors and the state of physical chemistry?
- What is the relationship between the type of institution and the state of physical chemistry?

A SURVEY WAS DEVELOPED TO MEASURE:

- Depth and breadth of content
- How content is delivered
- How student understanding is assessed
- Experiences and beliefs of instructors

SAMPLE:

- 331 instructors completed the survey
- 187 instructors provided contact information, which allows their response to be associated with their institution

RESULTS:

Description of the Institutions and the Instructors

Table 1. Description of the Institutions

Type of Institution	
Private	42
Public	58
Highest Degree	
Baccalaureate	28
Master's	34
Doctorate	38
Geographic Region	
West	13
South	27
Midwest	30
Northeast	30
Size	
Very Small	1
Small	30
Medium	32
Large	37
Setting	
Highly Residential	41
Primarily Residential	34
Primarily Nonresidential	25

Table 2. Description of the Instructors

Years of Teaching Experience	
1 to 5	16
6 to 10	18
11 to 15	20
15 to 20	12
At least 20	34
Area of Research	
Theoretical Chemistry	33
Experimental Chemistry	49
Both	18

Table 3. Description of Responsibilities

Teaching	
Total	55
Baccalaureate	65
Master's	65
Doctorate	42
Research	
Total	28
Baccalaureate	18
Master's	21
Doctorate	39
Other	
Total	17
Baccalaureate	17
Master's	14
Doctorate	19

RESULTS:

Description of Course Content, Delivery, and Assessment

Table 4. Thermodynamics Content Areas: Degree of Coverage and Type of Understanding

Content Area	Degree of Coverage			Type of Understanding		
	Great	Moderate	None	Both	Conceptual	Mathematical
Second Law of Thermodynamics	90	10	0	95	3	2
First Law of Thermodynamics	88	12	0	94	3	3
Helmholtz and Gibbs Energies	86	14	0	93	3	4
Chemical Equilibria	73	25	2	94	4	1
Third Law of Thermodynamics	65	35	0	82	16	2
Properties of Gases	63	36	1	93	4	3
Rate Laws	69	24	7	91	4	5
Phase Equilibria	57	40	3	74	24	2
Reaction Mechanisms	47	42	11	74	25	1
Kinetic Theory of Gases	46	44	10	76	21	3
Boltzmann Factor	36	54	10	76	18	6
Liquid – Liquid Solutions	35	53	12	69	30	1
Partition Functions	30	36	34	72	22	6
Solid – Liquid Solutions	21	57	22	63	36	1
Reaction Dynamics	19	52	29	52	46	2
Solids and Surface Chemistry	2	27	71	43	56	1

Table 5. Quantum Mechanics Content Areas: Degree of Coverage and Type of Understanding

Content Area	Degree of Coverage			Type of Understanding		
	Great	Moderate	None	Both	Conceptual	Mathematical
Particle in a Box	91	9	0	95	1	4
Harmonic Oscillator	86	14	0	89	7	4
Hydrogen Atom	84	15	1	89	7	4
Rigid Rotator	79	20	1	86	9	5
Postulates	67	32	1	70	27	3
Diatomic Molecules	66	31	3	73	25	2
Molecular Spectroscopy	52	43	5	69	28	3
Multi Electron Atoms	48	48	4	55	43	2
Free Particle	39	49	12	74	21	5
Approximation Methods	35	55	10	61	33	6
Polyatomic Molecules	28	59	13	50	50	0
History of Quantum Mechanics	14	78	8	33	66	1
Symmetry and Group Theory	17	36	47	60	38	2
Computational Methods	12	63	25	40	56	4
Laser Spectroscopy	10	49	41	37	63	0
Classical Mechanics	8	68	24	68	26	6

Table 6. Description of Course Delivery

Instructor-Centered	
Total	80
Baccalaureate	23
Master's	36
Doctorate	41
Equally Both	
Total	18
Baccalaureate	44
Master's	26
Doctorate	30
Student-Centered	
Total	2
Baccalaureate	80
Master's	20
Doctorate	0

Table 7. Description of Assessments

Type of Question	
Subjective	87
Objective	13
Type of Question	
Mathematical	62
Conceptual	38
Final Course Grade	
A	21
B	36
C	28
D	8
F	4
W	3

RESULTS:

Description of Instructor Beliefs

Table 8. Instructor Beliefs about Student Understanding

Students Struggle:	
Because they lack the necessary math background	61
To make connections between concepts and mathematics	31
To understand concepts	13
Because they do not put forth the necessary effort	18
Student Understanding May be Improved If:	
Instructors modify their teaching strategies	23
Students put forth more effort	20
Improved resources are available	10
More relevant examples and applications are available	8
Challenges of Developing New Teaching Strategies:	
Lack of time	89
Lack of resources	51
Lack of experience	39
Resistance from colleagues and department heads	22
Lack of interest	17

EMERGING THEMES:

- Institutions and Instructors
  - Large variation
- Responsibilities
  - Baccalaureate and master's institutions allocate more time for teaching responsibilities
  - Doctorate institutions allocate more time for research responsibilities
- Course Content
  - Cluster of commonly covered concepts
  - Cluster of concepts with large variation in coverage
  - Both a conceptual and mathematical understanding is preferred
- Course Delivery
  - Doctorate institutions have the greatest percent of instructor-centered teaching
  - Baccalaureate institutions had the greatest percent of student-centered teaching
- Assessment
  - Questions are primarily subjective, and assess both a conceptual and mathematical understanding.
  - 85% of students pass with an A, B, or C
- Instructor Beliefs
  - Students lack a strong math background
  - Instructors and students need to modify their behavior
  - Instructors require more time and resources